Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/04/2015
PI Name:	Judex, Stefan Ph.D.		
Project Title:	Genes that Predict the Loss of Bone during Weightlessness		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	stefan.judex@sunysb.edu	**Fax:**	FY 631-632-8577
PI Organization Type:	UNIVERSITY	**Phone:**	631-632-1549
Organization Name:	SUNY Stony Brook		
PI Address 1:	Dept. of Biomedical Engineering		
PI Address 2:	Bioengineering Building 213		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-5281	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	11/01/2012	**End Date:**	10/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
Contact Email:	elizabeth.taylor-23@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2016 per NSSC information (Ed., 9/10/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Donahue, Leah Rae (Jackson Laboratory)		
Grant/Contract No.:	NNX12AL25G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The amount of bone loss between individual astronauts is highly variable with some astronauts losing large amounts of bone while others are largely unaffected. This variability is accounted for, at least to a large extent, by genetic variations between individuals. Unfortunately, the specific genes that modulate this genetic sensitivity of bone loss in microgravity are unknown. To this end, we have recently completed a genetic study in which we identified specific regions on chromosomes (QTL or Quantitative Trait Loci) that either add to or reduce the propensity of an individual to lose bone when normal weightbearing is removed from the musculoskeleton. While providing critical information, these QTL still contain hundreds of genes, rather than a single one. Here, we propose first to greatly reduce the number of candidate genes with bioinformatics tools and then experimentally test the shortlist of candidate genes using knockout mice available from the Jackson Laboratory's (JAX) Genetic Resource Repository and the JAX Knock Out Mouse Project		

	(KOMP) production center. The identification of single genes modulating bone loss during weightlessness can be used as a critical diagnostic tool for the identification of astronauts most in need of countermeasures as well as to unravel novel drug targets that preserve bone health in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal directly addresses one of the six highest priority recommendations made in the National Research Council's 2011 Decadal Survey Report, "Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era" (http://www.nap.edu/), namely recommendation AH3: "Bone loss studies of genetically altered mice exposed to weightlessness are strongly recommended." Defining some of the specific genes that cause differences in the skeletal sensitivity to weightlessness will not only provide novel diagnostic and therapeutic means of combating bone loss in space but also on Earth in individuals subjected to bedrest or suffering from paralysis.
Task Progress:	Normal circulatory levels of estrogen are critical for maintaining bone health. Estrogen depletion post menopause or after surgery is detrimental to bone [1, 2]. Estrogen signals via two nuclear receptors: ESR1 and ESR2. ESR2 null mice reportedly overexpress ESR1 and core binding factor alpha 1 (cbfa1), a protein that promotes osteoblastic cell differentiation [3, 4]. Female ESR2 KO mice exhibit enhanced cortical bone sensitivity to mechanical signals [5]. We hypothesized that ESR null mice will exhibit an enhanced sensitivity to the removal of mechanical signals, leading to more pronounced losses in cortical bone when compared to wild type (WT) mice. Fas, a cell surface death receptor is expressed both in osteoblasts and osteoclasts [6, 7]. Fas deficiency protects mice from ovariectomy induced bone loss [7]. Hence, we hypothesized that Fas null mice will be spared from hindlimb unloading induced bone deterioration. Because some of the KO strains were bred on a B6N rather than a B6J background, we recently showed that even subtle genetic variations between C57BL/6J (B6J) and C57BL/6N (B6N) mice can alter bone morphology and response to hindlimb unloading. As detailed below, here, we report that these variations are due to differences in dynamic indices of bone formation, rather than bone resorption. To test the skeletal functions of ESR2 and Fas, ESR2 KO (ESR2-/-, n=20) and Fas KO (Fas-/-, n=20) were assigned to baseline control, age matched control, and hindlimb unloaded (HLU) groups. C57BL/6J (B6J, n=34) and C3H/HeJ (C3H, n=34) mice, the background strain of the ESR2 and Fas Kos respectively, were used as wild type (WT) controls. At the end of the experiment, the mice were sacrificed and femurs were harvested, fixed, and stored in 70% ethanol. The distal femoral metaphysis was subjected to microcomputed tomography (μCT) scanning and indices describing trabecular and cortical bone were quantified. The effect of gene deletion was tested by comparing age matched control KO mice to WT mice using two-tailed, independent samples T tests. The impact of HLU within a given genetic strain was tested for significance via one-way ANOVA and Tukey post-hoc test. Differences in magnitude of bone deterioration between WT and KO mice were tested by two-way ANOVA. ESR2 null mice had 17% greater bone volume and 7% more trabecular struts while trabecular thickness was 4% lower than WT-B6J mice at 11 weeks. ESR2 null mice had 11% greater cortical area but cortical thickness was similar to WT mice at 11 weeks (all $P<0.05$). ESR2 null mice were more susceptible to hindlimb unloading induced bone loss with 56%, 14%, and 27% losses in bone volume fraction (BV/TV), trabecular number (Tb.N), and trabecular thickness, while WT-B6J mice exhibited only 31%, 5%, and 19% lower BV/TV, Tb.N, and Tb.Th following HLU (Gene*HLU, $P<0.05$). Cortical response to HLU was similar between ESR2 null and WT mice. Fas deficiency did not alter trabecular or cortical bone morphology at 11 weeks of age. HLU Fas-/- mice had 49%, 21%, and 18% lower BV/TV, Tb.N, and Tb.Th when compared to their respective age matched control (all $P<0.05$) while WT-C3H mice had 53%, 26%, and 16% lower BV/TV, Tb.N, and Tb.Th (all $P<0.05$). However, no differences in the magnitude of the response were detected between Fas and WT-C3H mice for trabecular bone loss. Cortical bone responses were also similar between Fas and WT mice. We recently reported that trabecular architecture of B6N mice was more susceptible to HLU than in B6J mice. Consistent with this data we found that HLU B6N mice have a 29% and 21% lower bone formation rate (BFR/BS) and mineralizing surface (MS/BS) when compared to baseline control mice. In contrast, HLU did not suppress bone formation indices beyond the levels of baseline control in the B6J substrain (Gene*HLU, $P<0.05$). Osteoblast surface and osteoblast number were 59% and 51% lower in HLU B6N mice when compared to baseline control, while in the B6J substrain, these differences amounted to only 26% and 19% (Gene*HLU, $P<0.05$). No differences in osteoclastic activity were observed between the two substrains. These data highlight the role of ESR2 and Fas genes in maintaining bone health during physiological conditions and during reduced mechanical demand. Further, the differences that we observed between the two substrains, B6J and B6N, emphasize that differential levels of bone formation may have to be taken into account when using B6J and B6N. References 1. Riggs, B.L., The mechanisms of estrogen regulation of bone resorption. Journal of Clinical Investigation, 2000. 106(10): p. 1203. 2. Black, D.M., et al., The effects of parathyroid hormone and alendronate alone or in combination in postmenopausal osteoporosis. New England Journal of Medicine, 2003. 349(13): p. 1207-1215. 3. Windahl, S.H., et al., Female Estrogen Receptor ß-/- Mice Are Partially Protected Against Age-Related Trabecular Bone Loss. Journal of Bone and Mineral Research, 2001. 16(8): p. 1388-1398. 4. Nishimura, R., et al., Core-binding factor a 1 (Cbfa1) induces osteoblastic differentiation of C2C12 cells without interactions with Smad1 and Smad5. Bone, 2002. 31(2): p. 303-312. 5. Saxon, L., et al., Estrogen receptors a and ß have different gender-dependent effects on the adaptive responses to load bearing in cancellous and cortical bone. Endocrinology, 2012. 153(5): p. 2254-2266. 6. Katavic, V., et al., Increased bone mass is a part of the generalized lymphoproliferative disorder phenotype in the mouse. The Journal of Immunology, 2003. 170(3): p. 1540-1547. 7. Kovacic, N., et al., Fas receptor is required for estrogen deficiency-induced bone loss in mice. Laboratory investigation, 2010. 90(3): p. 402-413.

	8. Krishnan, V., H.U. Bryant, and O.A. MacDougald, Regulation of bone mass by Wnt signaling. Journal of Clinical Investigation, 2006. 116(5): p. 1202. 9. Spatz, J.M., et al., Sclerostin antibody inhibits skeletal deterioration due to reduced mechanical loading. Journal of Bone and Mineral Research, 2013. 28(4): p. 865-874. 10. Leupin, O., et al., Bone overgrowth-associated mutations in the LRP4 gene impair sclerostin facilitator function. Journal of Biological Chemistry, 2011. 286(22): p. 19489-19500.
Bibliography Type:	Description: (Last Updated: 01/16/2018)
Abstracts for Journals and Proceedings	Sankaran JS, Varshney M, Donahue LR, Judex S. "ESR1 and ESR2 Exert Opposing Influence on Bone's Susceptibility to Unloading." 37th Annual Meeting of the American Society for Bone and Mineral Research, Seattle, Washington, October 9-12, 2015. Journal of Bone and Mineral Research. 2015;30 (Suppl 1). , Oct-2015
Abstracts for Journals and Proceedings	Sankaran JS, Zhang W, Donahue LR, Judex S. "FGF2-/- and NOD2-/- mice have distinct contributions to bone health during unloading and under habitual loading conditions." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	Kuriakose S, Varshney M, Donahue LR, Judex S.. "CD44 Maintains Trabecular Compartment of Vertebral Body but Does Not Regulate Response to Disuse." 2014 Biomedical Engineering Society Annual Meeting, San Antonio, Texas, October 22-25, 2014. 2014 Biomedical Engineering Society Annual Meeting, San Antonio, Texas, October 22-25, 2014. , Oct-2014
Articles in Peer-reviewed Journals	Wallace IJ, Judex S, Demes B. "Effects of load-bearing exercise on skeletal structure and mechanics differ between outbred populations of mice." Bone. 2015 Mar;72:1-8. Epub 2014 Nov 22. http://dx.doi.org/10.1016/j.bone.2014.11.013 ; PubMed PMID: 25460574 , Mar-2015
Articles in Peer-reviewed Journals	Manske SL, Vijayaraghavan S, Tuthill A, Brutus O, Yang J, Gupta S, Judex S. "Extending rest between unloading cycles does not enhance bone's long-term recovery." Medicine & Science in Sports & Exercise. 2015 Feb 9. [Epub ahead of print] http://dx.doi.org/10.1249/MSS.0000000000000636 ; PubMed PMID: 25668404 , Feb-2015
Articles in Peer-reviewed Journals	Wallace IJ, Gupta S, Sankaran J, Demes B, Judex S. "Bone shaft bending strength index is unaffected by exercise and unloading in mice." Journal of Anatomy. 2015 Mar;226(3):224-8. http://dx.doi.org/10.1111/joa.12277 ; PubMed PMID: 25645569 , Mar-2015
Articles in Peer-reviewed Journals	Belavy DL, Adams M, Brisby H, Cagnie B, Danneels L, Fairbank J, Hargens AR, Judex S, Scheuring RA, Sovelius R, Urban J, van Dieën JH, Wilke HJ. "Disc herniations in astronauts: what causes them, and what does it tell us about herniation on Earth?" European Spine Journal. 2015 Apr 18. [Epub ahead of print] http://dx.doi.org/10.1007/s00586-015-3917-y ; PubMed PMID: 25893331 , Apr-2015
Articles in Peer-reviewed Journals	Wallace IJ, Pagnotti GM, Rubin-Sigler J, Naeher M, Copes LE, Judex S, Rubin CT, Demes B. "Focal enhancement of the skeleton to exercise correlates to mesenchymal stem cell responsivity rather than peak external forces." J Exp Biol. 2015 Jul 31. [Epub ahead of print] PubMed PMID: 26232415 , Jul-2015